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January 05, 2026

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Lane Change Prediction in Autonomous Driving: A Statistical Survey on Intention and Trajectory Forecasting

Mohamed Manzour, Rubén Izquierdo and Miguel Ángel Sotelo

***Abstract*—Lane-change prediction has emerged as one of the core challenges in autonomous driving research over the past decade. Previous surveys have focused on either lane-change intention or trajectory prediction, without offering a unified view of both tasks. They provide limited statistical analysis, offering mostly qualitative insights without capturing research trends. Additionally, they overlook interpretability and explainability, which crucial for trustworthy prediction systems. In this survey, we conduct a comprehensive review of over 150 research articles to systematically analyze how the field has evolved across different key dimensions: prediction tasks (intention, trajectory, or both), types of sensors used, considered inputs, datasets employed, modeling approaches, and evaluation practices. Beyond summarizing existing efforts, this survey also presents a statistical analysis that exposes critical gaps in the literature. The analysis reveals a lack of focus on interpretability, transparency, and explainability of the models, despite their importance for building reliable and trustworthy prediction systems. Additionally, only a small subset of studies incorporate decision-making modules that act upon the predicted behaviors, and even fewer offer experimental or real-world validation. Based on our findings, we outline recommendations for future research to advance toward more explainable, decision-aware, and deployable lane-change prediction systems.**

***Index Terms*—Lane Change Prediction Survey, Intention Prediction, Trajectory Forecasting, Autonomous Driving, Explainable AI**

I. Introduction

LANE change maneuvers are among the most frequent and critical behaviors in road traffic, especially in highway and urban merging scenarios. According to [1], 60–70% of traffic accidents result from vehicle collisions, it was highlighted that unsafe lane changes are the major contributing factor. Similarly, [2] reports that 75% of traffic accidents stem from drivers' errors during lane-changing maneuvers. These findings emphasize the importance of predicting lane changes as a critical function in Advanced Driver-Assistance Systems (ADAS) and in autonomous vehicles. Such predictive capabilities can enhance safety by helping the vehicle anticipate the intentions of surrounding vehicles or by forecasting the driver's own future maneuvers and communicating them to nearby traffic. To address this challenge, recent research has increasingly focused on two key tasks: intention prediction and trajectory prediction. Intention prediction refers to classifying the discrete future action that will be taken by the vehicle, such as keeping the lane or changing to an adjacent lane. Trajectory prediction, on the other hand, focuses on estimating the vehicle's future positions. Despite significant progress in both areas, the research landscape is still not well connected. Most of studies focus exclusively on either intention prediction or trajectory prediction, with relatively few addressing both tasks jointly. Furthermore, several survey articles have reviewed the literature in this domain, but most of them limit their scope to either trajectory prediction or intention prediction. Moreover, existing surveys often focus on describing methodologies without providing deeper statistical insights about past, current, and future research directions, such as the most commonly used models, sensors, or datasets.

In this article, we aim to bridge these gaps by providing a comprehensive survey that jointly covers lane change intention and trajectory prediction. This survey conducts a comprehensive review of over 150 research articles on lane-change prediction published between 2007 and 2025. Figure 1

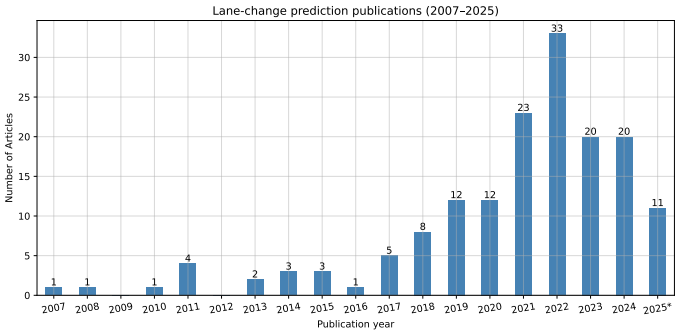

Fig. 1: Annual number of lane-change-prediction papers retrieved with Citation Chaser using the Lens.org bibliographic index. The graph shows the growth of research activity on lane change intention and trajectory prediction over almost two decades, resulting in more than 20 publications in peak years. Counts are based on the articles analysed in this review.

visualizes how publication volume has accelerated over that period, with more than 20 papers appearing in the peak years. The article then outlines the entire pipeline by providing an overview of the entire process, from sensors and input features to prediction methods and how the models are evaluated. Special attention is given to open discussions and ongoing challenges related to the interpretability, transparency, and

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*It is early to draw conclusions about 2025, as the year was still ongoing at the time this article was written.

explainability of current methods, as well as their integration with decision-making and real-world validation. A statistical analysis of the reviewed literature is also presented to identify trends and highlight critical research gaps. The rest of this article is structured as follows. Section II reviews previous surveys and presents our literature collection methodology. Section III analyzes the task distribution and trends across intention and trajectory prediction. Section IV describes the sensors and datasets used in the literature. Section V covers input features. Section VI discusses modeling approaches. Section VII presents evaluation methods, and Section VIII discusses open challenges regarding explainability, decision-making, validation, and future research directions. Finally, Section IX concludes article.

II. Survey Scope and Literature Collection

This section presents a brief review of existing survey articles that address lane change intention prediction and lane change trajectory prediction. The purpose of this overview is to clarify the scope and focus of previous surveys, identifying their primary contributions as well as the limitations in their coverage. These limitations serve as the motivation for the current survey, which aims to address the identified gaps and introducing a more comprehensive analysis. Additionally, the methodology used to collect and select the relevant scientific literature included in this survey is described.

A. Existing Surveys

The survey presented in [3] provided an overview of machine learning-based models for predicting lane-change behavior, with a specific focus on ego vehicles. The authors examined studies employing Artificial Neural Networks (ANN), Hidden Markov Models (HMM), and Support Vector Machines (SVM). They discussed various input variables that influence these models, such as environmental inputs (e.g. vehicle speed, acceleration, and heading), eye-tracking, head-tracking, and physiological data. A key limitation of this survey is its narrow methodological scope, as it considered only machine learning approaches and excluded alternative model families, such as probabilistic or knowledge-based methods. Moreover, its focus was limited to intention prediction for ego vehicles, without addressing surrounding vehicles or trajectory predictions. Another survey presented in [4] focused on trajectory prediction, specifically for surrounding vehicles, across a variety of driving scenarios such as lane changes, intersections, and merging situations. Unlike studies limited to a single modeling approach, this survey categorized the reviewed literature into four main groups: physics-based models, classical Machine Learning (ML) models, Deep Learning (DL) methods, and Reinforcement Learning (RL) approaches. For each article, the authors provided information on the specific model used, the corresponding evaluation metrics, and the reported results. Additionally, the survey identified the contextual input features employed by each article, offering a comprehensive view of the data sources used across the literature. Although this survey addressed a broader range of scenarios and modeling techniques, it did not consider important model-related aspects such as understandability or interpretability from the perspective of human users, which is currently an important aspect for a trustworthy model in safety-critical applications. The survey presented in [5] focused on lane change intention prediction for ego vehicles in highway scenarios. The authors reviewed a variety of ML model families, including generative models such as Dynamic Bayesian Networks (DBN) and HMM, discriminative models like SVM and Logistic Regression, as well as DL approaches such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks. For each reviewed study, the survey reported the input features used, the learning algorithm applied, and whether the data were collected in simulation environments or real-world settings. Additionally, the authors considered both the performance metrics and the prediction horizons associated with each model. Despite providing a structured comparison of algorithms and inputs for ego vehicle intention prediction, this survey did not extend its scope to include the prediction of surrounding vehicle intentions. Additionally, it lacked an analysis of whether the reviewed models were integrated into downstream decision-making modules or validated through simulation and hardware-in-the-loop experiments. A different survey presented in [6] focused on lane change intention prediction for surrounding vehicles. The authors began by analyzing the types of sensors used for data collection across various datasets. They provided insights into the cost-effectiveness and range capabilities of each sensor type, distinguishing between short-range and long-range sensing technologies. The survey then categorized the reviewed studies according to the modeling approach adopted, including probabilistic methods and machine learning-based models. For each study, the authors reported the dataset used, such as public datasets like Next Generation Simulation (NGSIM), highD, or the PREdiction of VEhicles iNTentIONs (PREVENTION) dataset, or indicated whether the models were trained using privately collected on-road data. They also detailed the types of input features employed, including vehicle status variables (e.g., position, velocity, acceleration), road geometry, traffic conditions, and weather information. However, the survey did not explore whether the reviewed models were integrated into decision-making frameworks or validated through hardware-in-the-loop experiments. Furthermore, it did not address critical aspects such as model interpretability, explainability, or understandability, which are essential for ensuring user trust in real-world applications.

Reviewing the existing surveys reveals several important gaps. Most of these surveys follow a paper-by-paper approach, presenting large tables that describe the datasets, input features, and models used in each study. However, they do not provide an overall analysis that looks across all the reviewed works at once. There are no figures or statistical summaries that help the reader understand general trends in the field. As a result, the reader is left with a list of individual contributions, but without a clear understanding of how the field is evolving or where the main focus areas lie. Another limitation is the narrow scope of many surveys. Some concentrate only on intention prediction, while others deal exclusively with trajectory prediction. Similarly, some focus only on ego vehicles, and others only on surrounding vehicles.

This separation makes it difficult for readers to compare these tasks or understand how methods, input features, and datasets usage differ between them. Furthermore, important aspects such as model interpretability and explainability are not discussed, even though some individual papers do address these topics. In addition, the surveys did not indicate whether the studied models are used in decision-making systems or validated through real-world hardware experiments. These are critical points for assessing how practical and trustworthy these models are in real-world driving scenarios.

B. Objective and Contribution

Based on the limitations identified in existing surveys, the objective of this survey is to address those gaps through the following contributions:

- **Comprehensive statistical analysis:** A detailed quantitative study is conducted to uncover how research efforts are distributed across prediction tasks, vehicle perspectives, input features, model types, and validation settings. This analysis reveals key trends, and potential directions for future work, going beyond a descriptive review to offer data-driven insights.
- **Task coverage:** Both intention and trajectory prediction are reviewed and compared in terms of commonly used datasets, input features, modeling approaches, and evaluation strategies.
- **Vehicle focus analysis:** A clear distinction is made between studies focused on the ego vehicle and those focused on surrounding vehicles, examining how research varies in terms of: input features, dataset selection, and modeling approaches.
- **Model diversity:** All categories of methods are included (ranging from traditional machine learning to deep learning and hybrid models) without restricting the analysis to a specific model family.
- **Model understanding:** The survey explicitly evaluates whether the reviewed works consider aspects such as interpretability, transparency, and explainability, which are rarely discussed in previous surveys.
- **Validation setting coverage:** Each study is categorized based on whether it is validated on datasets only, simulation environments, or real-world hardware.

C. Literature Collection Method

All articles reviewed in this survey were collected using the snowballing method [7]. The process begins with an initial set of seed papers directly related to lane change prediction [8–14]. These seed papers were selected to represent a range of publication years, various methodologies, prediction tasks (intention prediction, trajectory prediction, or both), and validation environments (e.g., simulation, real-world experiments, or numerical evaluation), in order to ensure a diverse and representative starting point for the snowballing process. These papers were then used as input to a tool called Citation Chaser [15] which is based on the Lens.org bibliographic index, which automatically retrieves both backward citations (i.e., the references cited by the seed papers) and forward citations (i.e., papers that cited the seed papers). After collecting these citations, duplicate entries resulted from overlaps between backward and forward citations were removed. Next, the titles of the remaining papers were screened based on the presence of relevant keywords which are (lane change AND (prediction OR intention OR trajectory)). Papers whose titles matched the scope of the survey were further reviewed at the abstract and full-text levels. Only studies that clearly addressed lane change prediction (whether intention-based or trajectory-based) were considered. Irrelevant papers were excluded. Once the first iteration was complete, the newly accepted papers became the input for the next iteration of snowballing. The same process was repeated: backward and forward citations were collected using Citation Chaser, duplicates from previously reviewed papers were removed, and newly identified papers were screened for relevance. This iterative procedure continued until no additional relevant articles were found in the last iteration. In this survey, the snowballing process was carried out for four iterations after feeding the initial set. Figure 2 illustrates the complete flow of this process.

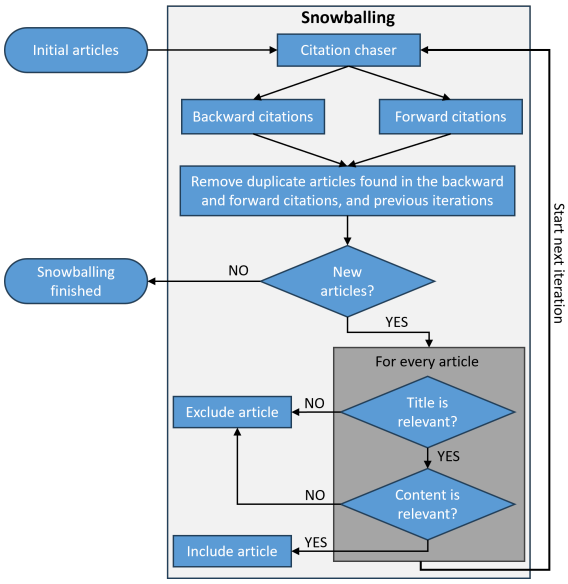

Fig. 2: Flowchart illustrating the snowballing methodology used for literature collection in this survey. Starting from an initial set of relevant articles, forward and backward citation chasing was performed using the Citation Chaser tool. Duplicates across citations and iterations were removed, and newly identified papers were assessed for relevance based on title and content screening. The iterative process continued until no additional relevant articles were found, marking the completion of the snowballing process.

III. Taxonomy of Prediction Tasks

A. Prediction Tasks

Over the past two decades, various studies have addressed the prediction of lane change maneuvers, with some focusing exclusively on intention prediction (e.g., [8–10, 16–19]), and others concentrating solely on trajectory prediction (e.g., [20–26]). In addition, several works have explored both intention

and trajectory prediction together. In such cases, some approaches first predict the driver's intention and then use the predicted intention as an input to a subsequent model that forecasts the trajectory, as seen in [11, 27–48]. On the contrary, other studies adopt the opposite strategy by predicting the trajectory first and using it to infer the underlying intention, as done in [49, 50]. There are also models that predict both intention and trajectory simultaneously using a single model where neither task is conditioned on the other; such examples include [51–56]. Figure 3 illustrates the annual distribution of publications on lane change prediction, categorized by the type of task addressed. From 2007 to 2016, the number of publications remained low, typically ranging from one to four per year. However, beginning in 2017, there was a noticeable increase in works focusing on intention prediction, peaking in 2022. Similarly, research on trajectory prediction began to rise significantly around 2020, maintaining a steady count through 2025. Nevertheless, it is early to draw conclusions about 2025, as the year is still ongoing.

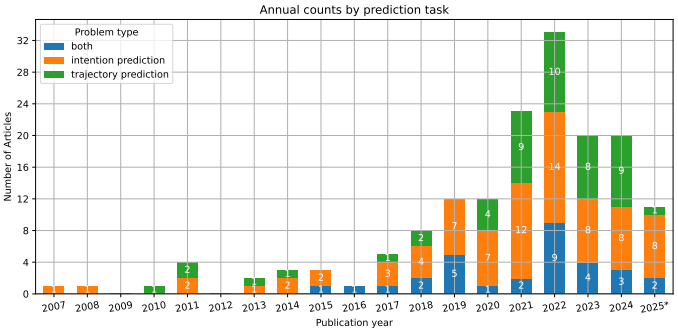

Fig. 3: Annual distribution of lane-change-prediction studies by problem type (2007 – 2025). Each stacked bar shows the number of papers published in a given year that address intention prediction only (orange), trajectory prediction only (green), or tackle both tasks in a single study (blue). Across the full interval, the survey includes: 80 articles (50%) focus solely on intention, 49 (30.6%) solely on trajectory, and 31 (19.4%) address both tasks.

B. Vehicle Focus

With regard to the vehicle of interest in prediction tasks, some studies focus on predicting the intentions of the ego vehicle (which is the vehicle equipped with the decision-making system) while others focus on predicting the intentions of the surrounding vehicles. Most studies focus on predicting the behavior of surrounding vehicles rather than the ego vehicle. As shown in Figure 4, after 2016 there was a notable increase in surrounding vehicle studies. However, research focused on ego vehicle remains a minority, never exceeding five papers in any year. Also, Figure 5 illustrates the distribution of these studies by vehicle focus and task type. It can be observed that most works targeting the ego vehicle behavior are limited to intention prediction only. Among the studies focused on intention prediction, approximately one-third (36%) address the ego vehicle, while the remaining two-thirds (64%) focus on surrounding vehicles. In contrast, when it comes to trajectory prediction or joint prediction of both intentions and trajectories, the focus shifts a lot toward surrounding vehicles. Very few studies address trajectory prediction or joint prediction for the ego vehicle as shown in the two rightmost pies. Looking to the leftmost pie chart of the figure, which aggregates all studies involving intention prediction, trajectory prediction, or both, only about one-fifth (22%) of the works focus on predicting the behavior of the ego vehicle. The remaining 78% focus on surrounding vehicles. Table I provides the detailed distribution of the analyzed articles based on prediction task and vehicle focus. The highest proportions are observed in studies that predict either the intentions or the trajectories of surrounding vehicles, comprising 31.9% and 28.1%, respectively. Studies addressing both intentions and trajectories for surrounding vehicles account for 18.1%. In contrast, studies focused on the ego vehicle's intention prediction represent 18.1%, while those predicting ego vehicle trajectories or both tasks jointly for the ego vehicle are rare, making up only 2.5% and 1.3%, respectively.

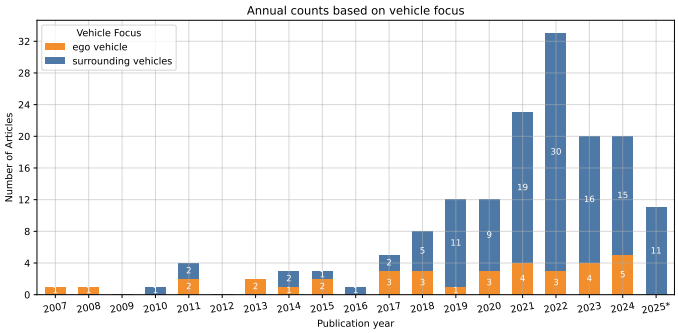

Fig. 4: Annual distribution of lane change prediction studies by vehicle focus. Across the reviewed articles, research focused on surrounding vehicles (blue) clearly dominates over ego vehicle research (orange) and its growth accelerates after 2016.

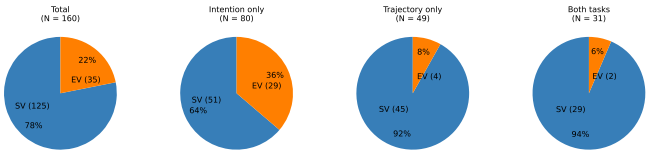

Fig. 5: Distribution of vehicle focus (surrounding vehicles—SV (blue) vs. ego vehicle—EV (orange)) across reviewed studies, grouped by prediction task. The four pie charts represent all tasks combined, intention prediction only, trajectory prediction only, and studies addressing both tasks. Overall, the majority of studies (78%) focus on surrounding vehicles, especially in trajectory-only (92%) and joint-task (94%) scenarios, whereas intention-only research shows a higher proportion of ego-vehicle focus (36%).

IV. Datasets and Sensors

This section presents the key datasets and sensor modalities used in lane change intention and trajectory prediction research. It is divided into two parts: one focusing on public and private datasets and their usage patterns, and another providing insights into the sensors commonly adopted in privately collected datasets.

TABLE I: Distribution of the reviewed studies by prediction task and vehicle focus. Research is heavily skewed toward surrounding vehicle (SV) behavior: 31.9% of papers predict SV intentions and 28.1% predict SV trajectories. In contrast, ego vehicle (EV) trajectory studies are rare (2.5%), and only 1.3% tackle joint tasks for the EV.

Prediction Task	Focus Vehicle	Article ID	Number of Articles
intention	EV	[8, 16, 17, 57–82]	29 (18.1%)
intention	SV	[9, 10, 13, 14, 18, 19, 83–127]	51 (31.9%)
trajectory	EV	[12, 20, 128, 129]	4 (2.5%)
trajectory	SV	[21–26, 130–168]	45 (28.1%)
both	EV	[41, 49]	2 (1.3%)
both	SV	[11, 27–40, 42–48, 50–56]	29 (18.1%)

A. Datasets

To develop models capable of accurately predicting vehicle intentions or trajectories, access to representative and realistic driving datasets is essential. These datasets can be broadly categorized based on their collection perspective. Some are gathered from the ego vehicle's viewpoint using onboard sensors such as the PREVENTION dataset [169], while others are collected from an external third perspective of view that provides a bird's-eye view of traffic scenes, such as the NGSIM [170] and highD [171] datasets. In addition to these publicly available datasets, several studies utilize their own privately collected datasets which are recorded using custom sensor configurations. This subsection provides a brief overview of the three mentioned widely used public datasets, followed by a statistical summary of dataset usage trends across different prediction tasks and vehicle focus.

1) NGSIM: The NGSIM dataset [170], developed in 2005 by the U.S. Department of Transportation, contains real-world vehicle trajectories from two freeway sections: US-101 and I-80. Data was collected using multiple overhead cameras mounted on buildings, capturing dense highway traffic over three 15-minute segments. The dataset includes trajectories for approximately 5,000 vehicles recorded at 10 Hz, along five main lanes and an additional on-ramp/off-ramp lane. NGSIM tracks vehicle positions, velocities, accelerations, and lane IDs, allowing for the reconstruction of lane change events by analyzing lane transitions over time. The dataset is publicly available and can be accessed at: https://ops.fhwa.dot.gov/trafficanalysistools/ngsim.htm.

2) highD: The highD dataset [171] (recorded in 2018) is a German highway dataset that captures naturalistic top-view traffic scenes using a camera mounted in a drone. The recordings were conducted across six different highway locations in Germany, resulting in 60 video tracks of approximately 15 minutes each and covering more than 110,500 vehicles. The data is recorded at a high frame rate (25 Hz) and provides comprehensive information for each vehicle, including position, speed, acceleration, and the current lane, as well as spatial relations with surrounding vehicles such as the existence of a preceding, following, or adjacent vehicle in each lane. Additionally, the dataset includes time-to-collision (TTC) and THW with the front vehicle. Researchers frequently derive lane keeping, left lane change, and right lane change labels by tracking lane ID transitions across time. The dataset is publicly available and can be accessed at: https://levelxdata.com/highd-dataset/.

3) PREVENTION: The PREVENTION dataset [169], released in 2019, was collected at a frequency of 10 Hz using an instrumented ego-vehicle equipped with six synchronized sensors, including Light Detection and Ranging (LiDAR), Radio Detection and Ranging (Radar), cameras, Global Positioning System (GPS), Inertial Measurement Unit (IMU), and Controller Area Network (CAN) bus. It contains 356 minutes of naturalistic highway and urban driving data across approximately 540 km in the Madrid region, recorded using three drivers. The dataset focuses on the surrounding vehicles from the perspective of the ego vehicle, providing over 3,000 annotated trajectories and more than 900 manually labeled lane changes (left and right), along with critical events such as cut-ins, cut-outs, and hazardous maneuvers. Vehicles are tracked via multi-sensor fusion, ensuring high spatial accuracy, and lane markings are included to support lateral and longitudinal localization. PREVENTION is notable for its ego-vehicle viewpoint and the inclusion of Radar, which enhances robustness in poor visibility conditions. The dataset is publicly available at: https://prevention-dataset.uah.es/.

4) Dataset Usage Trends: Figure 6 displays a heatmap ranking the datasets used in this survey according to their frequency of appearance, ordered from the most used on the left to least used on the right. The NGSIM dataset is the most frequently adopted, appearing in numerous works such as [9, 10, 19], and is predominantly used for trajectory prediction of surrounding vehicles. Since it is collected from a third-person perspective using fixed overhead cameras most studies assume the availability of complete vehicle state information by assuming the availability of Vehicle-to-Vehicle (V2V) communication. This assumption enables accurate prediction of surrounding vehicle motion. The second most used dataset is the highD dataset which has also been widely used, as in [11, 18, 21], and similarly focuses on surrounding vehicle behavior. Unlike NGSIM, highD is mostly used in intention prediction tasks. Both datasets enable inference of maneuvers by tracking lane ID transitions over time, but their third-person collection perspective limits their applicability for predicting the ego vehicle's intent because no access to the driver states and some vehicles states like the steering angle. In third place, a large portion of works use privately collected datasets, mostly designed for intention prediction tasks. Specifically, 21 studies focus on ego vehicle intentions, while only 9 target surrounding vehicle intentions. A small number of these datasets is applied to trajectory prediction tasks. Then, for the PREVENTION dataset, it is used to predict surrounding

vehicle intentions from the ego vehicle's perspective. Only one article focused on trajectory prediction of surrounding vehicles among 8 articles, the other 7 articles focused on predicting surrounding vehicle intentions. Additionally, Table II lists the four most frequently used datasets (NGSIM, highD, privately collected data, and PREVENTION), alongside the corresponding articles that used each one. Other datasets appear less frequently. For example, CitySim [172] [54, 69, 95], and CARLA Risky-lane-change Anticipation in Simulated Highways (CRASH) dataset [13, 14], have been used in a small number of studies. Datasets like Apollo [173], Peking University (PKU), University of Alcala (UAH), Stanford Drone Dataset (SDD) [174], and Strategic Highway Research Program (SHRP2) [175] each appear only once in the literature [49, 138, 152, 155].

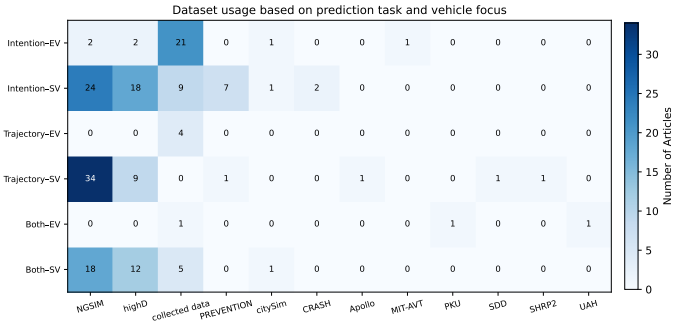

Fig. 6: Distribution of datasets used across lane-change prediction tasks (Intention, Trajectory, and Both) and vehicle focus. Cells indicate the number of papers utilizing each dataset for the corresponding prediction task and vehicle perspective. "Collected Data" refers to studies that utilized their own custom-collected datasets. NGSIM and highD datasets are the most widely used, particularly for tasks targeting surrounding vehicles. NGSIM dominates in trajectory prediction tasks focused on surrounding vehicles.

Figure 7 confirms the findings from Figure 6 by showing the annual usage trends of the four major dataset categories: NGSIM, highD, PREVENTION, and privately collected datasets. Before 2016, the use of datasets for prediction tasks was relatively limited, with NGSIM and a few private datasets being the only sources used in early studies. Starting from 2018, a noticeable increase in the usage of NGSIM and privately collected data can be observed. Articles that use the highD and PREVENTION started to increase from 2020, after their publication (in 2018 and 2019 respectively). A strong correlation can be observed between the increased usage of NGSIM and highD datasets and the growing number of studies focused on surrounding vehicles, particularly from 2020 onward. On the other hand, the trend line associated with private datasets (shown in red) demonstrates a clear rise in parallel with the increasing number of publications that focus on ego vehicle intention prediction. This suggests a direct relationship between the availability of privately collected data and the growth of studies focusing on the ego vehicle.

Based on this taken analysis, a major gap in the field is the lack of a public benchmark dataset that enables intention

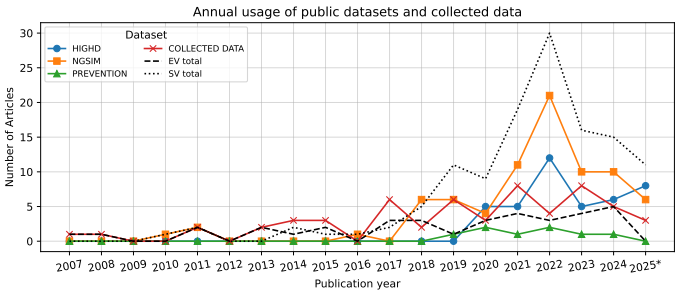

Fig. 7: Annual usage of major datasets and collected data, with total papers studying ego vehicles (EV, dashed line) versus surrounding vehicles (SV, dotted line). Peaks in the usage of NGSIM and highD closely track the rapid increase in studies focused on surrounding vehicles, while ego-vehicle research grows steadily, largely relying on collected data due to the lack of suitable public datasets.

prediction for the ego vehicle using internal cues, such as facial expressions, eye gaze, head pose, or biometric signals. Existing public datasets do not capture such internal signals, which are critical for understanding the driver's decision-making process. So, the development of a shared benchmark dataset combining external traffic context and driver-internal data would be highly valuable, enabling comparison of articles' results in the area of ego-vehicle intention prediction research.

B. Sensors

An overview of the sensors employed in privately collected datasets, including statistics on their frequency of use, categorized by both prediction task and vehicle focus is provided in Figure 8 as a heatmap of sensor occurrence across the surveyed articles. The most frequently used sensor is the

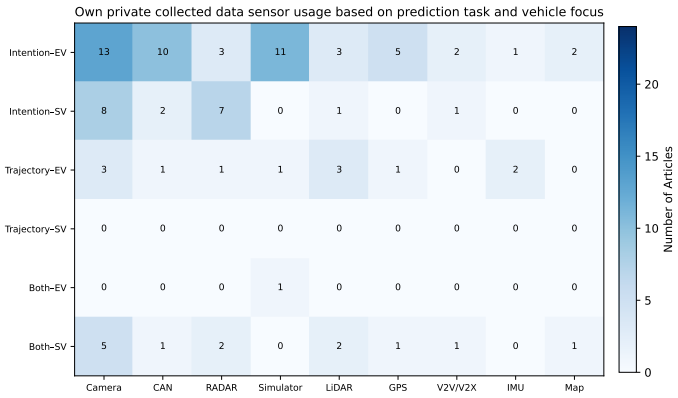

Fig. 8: Distribution of sensors utilized in studies with custom-collected data across prediction tasks and vehicle focus. Each cell represents the count of papers employing a specific sensor for the corresponding prediction task and vehicle perspective. Cameras and CAN-bus data are the predominant sensor choices for intention prediction, particularly when predicting the ego vehicle's lane-change maneuvers.

camera, appearing in a wide variety of setups. In some cases [8, 17, 61, 67, 71, 72], cameras are mounted inside the vehicle to monitor the driver's head pose and eye gaze, while

TABLE II: Overview of the most used datasets across reviewed studies. NGSIM and highD are the most frequently used public datasets, primarily for surrounding vehicle prediction.

Dataset	Article ID	Number of Articles
NGSIM	[9–11, 19, 21–27, 29, 30, 32–34, 36–40, 43, 45, 48, 52, 53, 56, 58, 59, 84, 86, 89, 90, 93–99, 104, 107–109, 115, 119, 122–125, 134–137, 139, 140, 142–160, 163–165]	78 (43.8%)
HighD	[11, 13, 14, 18, 21, 29, 33, 34, 38, 40, 42, 44, 47, 50, 55, 56, 70, 75, 92, 93, 95, 101, 102, 104–108, 113, 117–119, 126, 130–132, 134, 135, 141, 159, 162]	41 (23.0%)
Private Collected Data	[8, 12, 17, 20, 28, 31, 35, 41, 46, 51, 57, 60–68, 71–74, 77–80, 82–85, 91, 114, 116, 120, 121, 127–129]	40 (22.5%)
PREVENTION	[87, 88, 100, 103, 110–112, 133]	8 (4.5%)
Other Datasets	[13, 14, 49, 54, 69, 81, 95, 138, 152, 155]	11 (6.2%)

in other cases, they are positioned to face the road from different angles to observe the surrounding vehicles and the surrounding scene. The second most commonly used sensor is the CAN bus, which provides internal vehicle data such as speed, acceleration, steering angle, and steering rate. Other commonly used sources of data include LiDAR, Radar, GPS, IMU, HD maps, and communication modules for V2V/V2X. A few datasets also rely on simulation platforms for data generation. From the figure, it can be observed that the use of cameras and CAN bus data is strongly associated with studies focusing on ego vehicle intention prediction, aligning with earlier findings that private datasets focuses on predicting the driver behavior. These sensors enable to capture both vehicle dynamics and driver state cues, which are critical for modeling the decision-making/prediction process of human drivers in real-world scenarios.

V. Input Features

This section provides a detailed analysis of the input features used in lane change prediction models, examining their relevance to different prediction tasks (intention or trajectory), vehicle focus (ego or surrounding), input format (e.g., numerical or linguistic) feature domain (kinematic or contextual), and the length of input sequences.

A. Feature Usage Frequency and Relevance

Input feature selection varies according to the prediction task and the vehicle of interest. For ego-vehicle intention prediction, driver cues such as head pose, eye gaze, and driving wheel steering angle are sometimes considered. In contrast, predicting the behavior of surrounding vehicles often relies on external measurements such as relative distance and relative velocity to the other vehicles. Figure 9 presents a histogram of the most frequently used features across all surveyed studies. Velocity appears as the most dominant feature, used in 106 articles, followed by position (91), acceleration (73), relative distance (63), and relative velocity (46). Infrequently used features include head pose (7) and eye gaze (5), which are mostly applied in studies focusing on the ego vehicle. A prediction-task and vehicle-focus input breakdown is shown in the heat-map in Figure 10. Position is primarily used in trajectory prediction for surrounding vehicles, reflecting its importance in reconstructing and forecasting motion over time. Velocity and acceleration are widely employed across both tasks but are especially prominent in intention prediction. Additional features such as steering angle, yaw rate, head pose, and eye gaze appear in a small number of studies and are typically associated with intention prediction for ego vehicles.

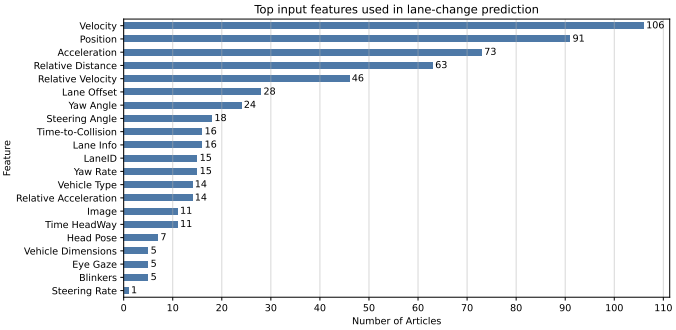

Fig. 9: Top input features used across all studies (count of individual feature usages). Basic kinematic features, such as velocity (106 papers), position (91), and acceleration (73), dominate the literature, while context cues (e.g., lane offset (28), head pose (7), blinkers (5)) are less frequently utilized.

B. Feature Domains: Kinematic vs. Contextual

Input features can be broadly categorized into kinematic and contextual domains. Kinematic features are directly related to vehicle motion, including basic features obtained from on-board sensors or computed using CAN bus data such as position, velocity, acceleration, yaw angle, steering angle, and derivative features which are obtained from these basic features such as relative distance, reltive velocity, steering rate, yaw rate, Time-to-Collision (TTC) and Time Headway (THW). Contextual features describe aspects of the environment or vehicle that are not derived from motion. These include lane ID, lane offset, lane info (lane curvature, number of lanes, lane markings style), vehicle type, blinkers, head pose, eye gaze, and vehicle dimensions. Figure 11 illustrates the distribution of input domains by prediction task. Trajectory prediction models rely on kinematic features in 69% of the cases, while 59% of intention prediction studies combine kinematic features with at least one contextual feature. In joint prediction tasks, 45% of the models incorporate both types. Figure 12 further decomposes this usage by prediction task and vehicle focus. Intention prediction for surrounding vehicles often includes contextual features (30 articles), while 21 studies rely only on kinematic data. Intention prediction for the ego vehicle also tends to emphasize contextual cues, particularly driver-related inputs. Trajectory prediction models, whether for the ego or surrounding vehicles, remain predominantly kinematics-driven, with minimal contextual integration. These patterns suggest a notable research gap in trajectory forecasting, where contextual features such as lane offset is rarely explored. Table III groups the surveyed papers by

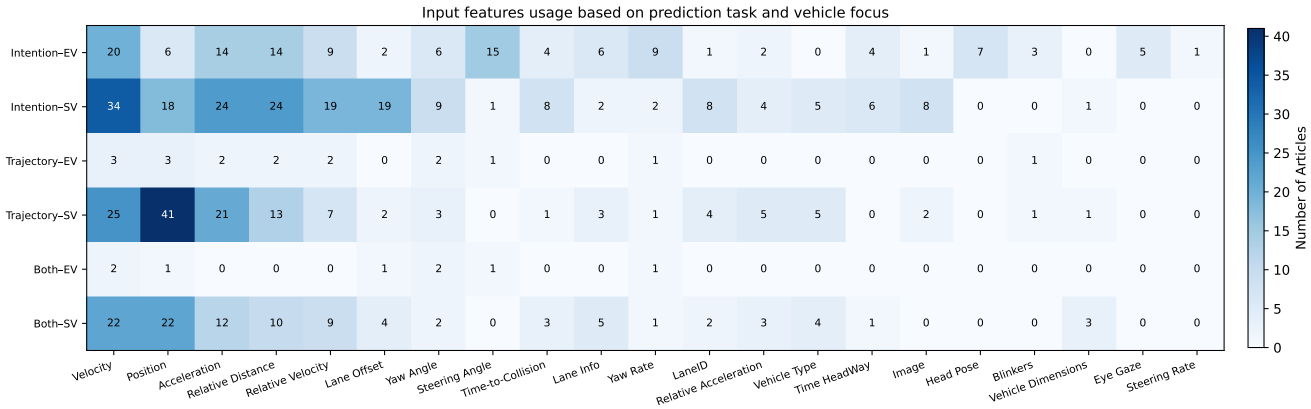

Fig. 10: Heat-map of feature usage split by prediction task and vehicle focus (Ego—EV / Surrounding—SV). Trajectory-SV models rely most heavily on raw kinematics, specifically position (41 papers) and velocity (25). Intention-SV models draw on richer contextual cues compared with Trajectory-SV, notably lane offset (19 papers). Intention-EV studies frequently include steering angle (15 papers). Features such as blinkers and eye-gaze are rarely used across all categories.

TABLE III: Input feature domain across prediction tasks and vehicle focus. Contextual inputs are more used in intention prediction tasks. However, kinematic features are still widely used, particularly for trajectory prediction tasks.

Prediction Task	Input Domain	Article ID	Number of Articles
intention	Kin.	[9, 10, 16, 18, 58–60, 62, 64, 66, 69, 70, 75, 76, 82, 88, 89, 91, 93, 94, 97–100, 104, 105, 113, 115, 116, 119, 120, 122, 126]	33 (20.6%)
intention	Con.	[8, 13, 14, 17, 19, 57, 61, 63, 65, 67, 68, 71–74, 77–81, 83–87, 90, 92, 95, 96, 101–103, 106–112, 114, 117, 118, 121, 123–125, 127]	47 (29.4%)
trajectory	Kin.	[12, 21, 23–26, 128–130, 135–138, 140, 143–147, 149, 151–153, 155–161, 163–166]	34 (21.3%)
trajectory	Con.	[20, 22, 131–134, 139, 141, 142, 148, 150, 154, 162, 167, 168]	15 (9.4%)
both	Kin.	[11, 29, 31, 32, 34, 38, 42–47, 49, 51–54]	17 (10.6%)
both	Con.	[27, 28, 30, 33, 35–37, 39–41, 48, 50, 55, 56]	14 (8.7%)

prediction task and input domain, presenting in each cell the corresponding article IDs, along with the count and percentage of studies. Figure 13 extends this snapshot by tracing how the use of kinematic and contextual features has changed over the years. While contextual inputs have become more common, kinematic features continue to play a key role in most studies.

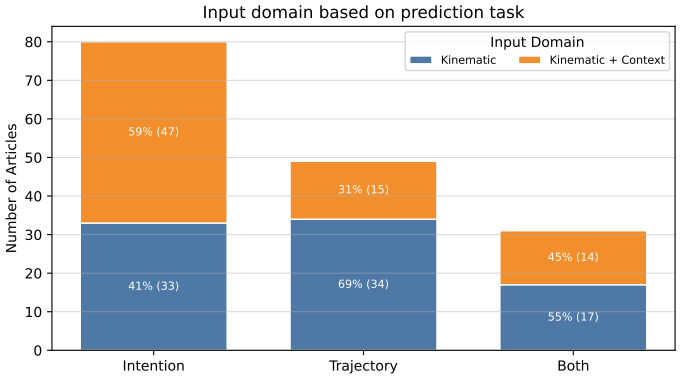

Fig. 11: Distribution of input domains (Kinematic vs. Kinematic + Context) across different prediction tasks. Intention models rely more on contextual cues; 59% of papers combine at least one contextual feature with other kinematical features. In contrast, Trajectory models remain mostly based on kinematical inputs (69%). Joint (Both) studies sit in between, with 45% using mixed inputs.

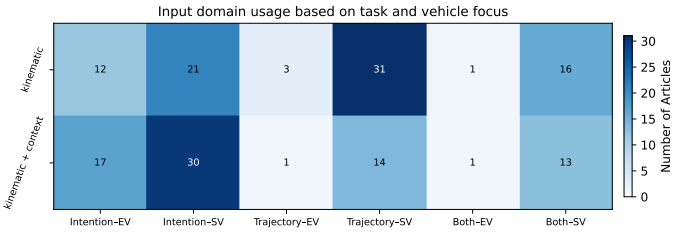

Fig. 12: Input domain usage based on prediction task and vehicle focus. Kinematic inputs dominate across most categories, especially for trajectory prediction focused on surrounding vehicles (31 papers) and ego vehicles (3). However, contextual features are more frequently incorporated in intention prediction, particularly for surrounding vehicles (30 papers vs. 21 for kinematic). This highlights a stronger tendency to include at least one contextual information when predicting the intentions of other vehicles.

C. Input Formats

Beyond feature content, the input format significantly influences model design. The majority of studies adopt numerical inputs as shown in Figure 14, but several alternative formats have been introduced:

- **Numerical inputs:** Feature values represented as continuous or discrete numbers; used in 96% and 84% of trajectory and intention prediction studies respectively.
- **Image-based inputs:** Employed in 11% of intention prediction studies, mainly for surrounding vehicles.

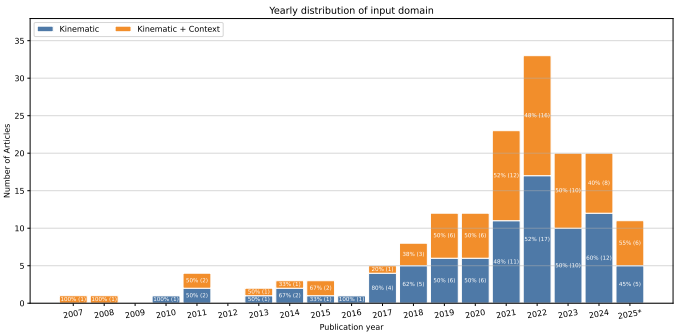

Fig. 13: Temporal distribution of input domains in lane change prediction studies (2007–2025). Each bar represents the proportion of studies published in a given year that rely solely on kinematic inputs (blue) versus those that incorporate both kinematic and contextual features (orange). From 2018 onward, both input strategies are used at nearly equal rates.

- **Linguistic inputs:** Encode vehicle behavior using natural language (e.g., "moving right" instead of numerical lateral velocity); used in 5% of intention prediction studies.
- **Prompt-based inputs:** Incorporate feature numerical values into natural-language prompts for large language models; only one study explored this format while predicting both intention and trajectory of lane changes.

Table IV provides a detailed listing of article IDs for image, linguistic, and prompt-based inputs. The limited use of linguistic and prompt-based formats reveals an opportunity to improve interpretability through more human-understandable feature representations.

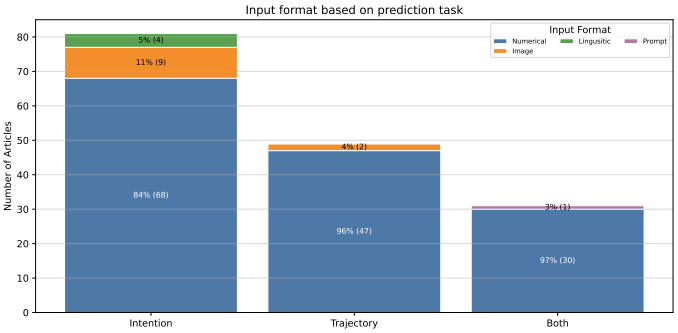

Fig. 14: Input-format distribution per prediction task. Numerical signals dominate every task, 96% of Trajectory studies and 97% of joint (Both) studies rely solely on numeric features. Intention studies are the only group that meaningfully explores alternatives, with 11% using image inputs and 5% employing linguistic representations. Prompt-style inputs appear in just one paper.

D. Input History Length

The input history length, which refers to the temporal window or sequence duration of past observations utilized by prediction models, is a critical factor in lane change prediction. Different studies have employed varying lengths of historical data, depending on the prediction task, and vehicle of interest.

TABLE IV: Summary of articles IDs that consider alternative input formats. While 11 articles adopt image-based inputs, only 4 use linguistic representations and a single article explores prompt-based inputs.

Dataset	Article ID
Image	[74, 85, 87, 103, 110–112, 118, 127, 132, 133]
Linguistic	[13, 14, 18, 113]
Prompt	[55]

As shown in Figure 15, studies focusing on the ego vehicle intention usually consider an input history length of approximately 2 seconds. In contrast, studies targeting surrounding vehicle intention prediction exhibit a broader range, with input lengths varying between 0 to 4 seconds. An input length of 0 seconds indicates that the model relies solely on instantaneous sensor readings without incorporating historical information, whereas 1-4 seconds implies that a temporal context of past observations is included before model inference. For trajectory prediction and joint prediction tasks, the majority of studies utilize around 3 seconds of input history.

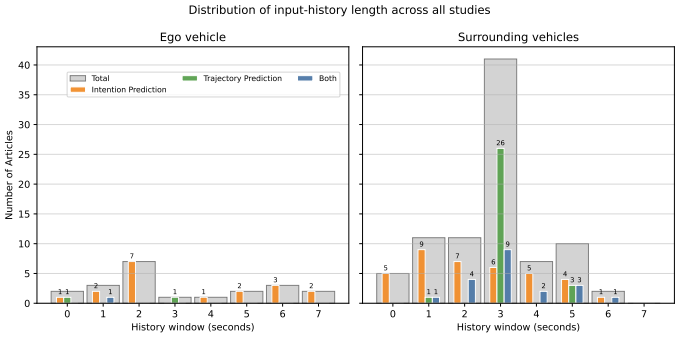

Fig. 15: Distribution of the input-history window lengths adopted across the surveyed lane-change-prediction studies. A clear preference for short temporal contexts $\leq$ 2s of past data is shown for ego vehicle studies. For surrounding vehicles, input length varies between 0s and 4s.

VI. Prediction Methods

This section presents a comprehensive overview of the modeling approaches used for lane change intention and trajectory prediction. We cover the full taxonomy of models, provide usage statistics and temporal trends, and analyze model usage by prediction task and vehicle focus.

A. Models Taxonomy

Various modeling approaches have been explored, spanning six main families: probabilistic and Bayesian models, RL, traditional ML, graph representation learning, DL, and physics-based models. Figure 16 illustrates this taxonomy.

1) Probabilistic and Bayesian family includes Bayesian networks (static, dynamic, and Naive Bayes) where dynamic ones include HMM, mixture models such as Gaussian Mixture Model (GMM) and Interactive Multiple Model (IMM), Gaussian process models, and Bayesian inference models.

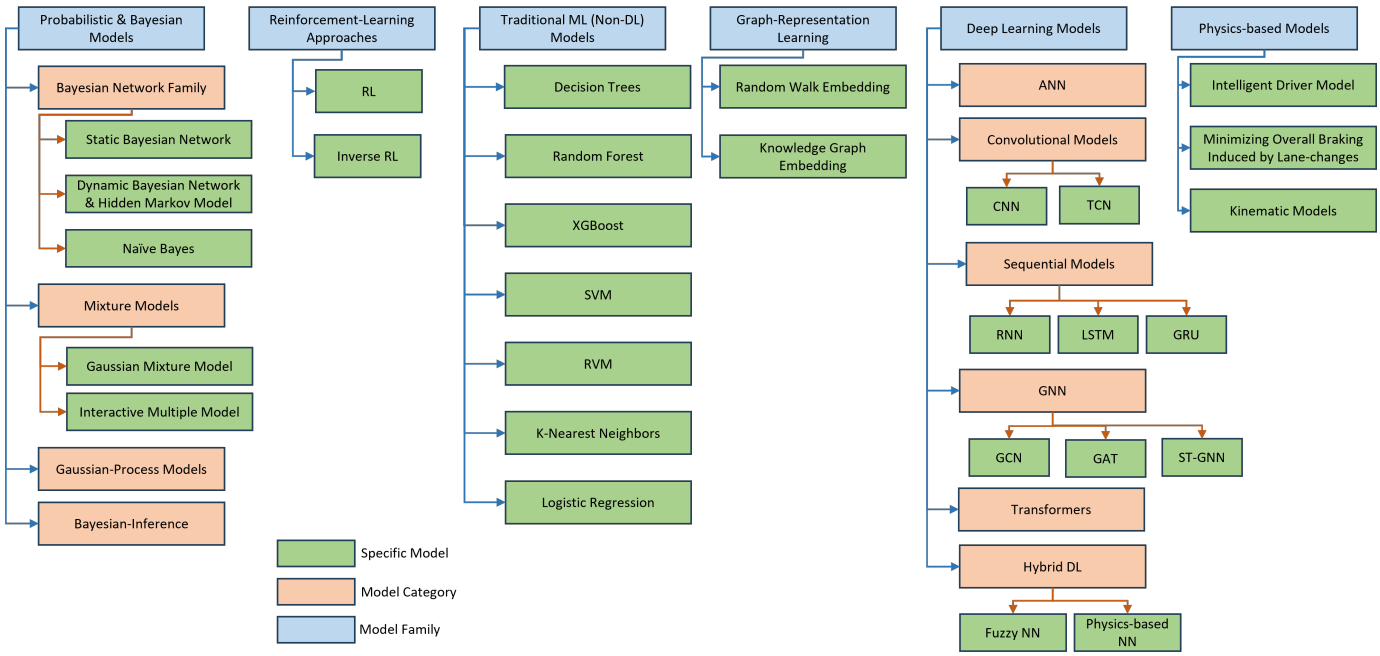

Fig. 16: Taxonomy of prediction models used in lane-change intention and trajectory prediction tasks. Models are grouped into six main families: Probabilistic & Bayesian, Reinforcement Learning, Traditional ML, Graph Representation Learning, Deep Learning, and Physics-based approaches. Color coding indicates hierarchy: blue boxes denote high-level model families, orange boxes represent subcategories or model classes, and green boxes indicate specific modeling techniques used in the literature.

2) RL approaches include both standard RL and inverse RL. These are used less frequently.
3) Traditional machine learning models remain common, including SVMs, Relevance Vector Machiness (RVMs), decision trees, random forests, eXtreme Gradient Boostings (XGBoosts), K-Nearest Neighborss (KNNs), and logistic regression. These are primarily used for intention prediction.
4) Graph representation learning models, particularly Knowledge Graph Embeddings (KGEs), have recently been introduced to incorporate semantic context and improve interpretability and transparency, often in combination with Bayesian inference.
5) DL models are the most widely adopted and are categorized as: ANNs, Convolutional models: Convolutional Neural Networks (CNNs) and Temporal Convolutional Networks (TCNs), Sequential models: RNNs, LSTMs, and Gated Recurrent Units (GRUs), Graph Neural Networks (GNNs): including Graph Convolutional Networks (GCNs), Generative Adversarial Networks (GANs), and Spatio-Temporal Graph Neural Networks (ST-GNNs), transformers, and hybrid models, combining neural networks with fuzzy logic or physics-based constraints.
6) Physics-based models include kinematic and rule-based approaches, such as the Intelligent Driver Model (IDM) and Minimizing Overall Braking Induced by Lane Changes (MOBIL) model.

B. Models Usage Trends

The frequency of model use across the surveyed literature is shown in Figure 17. Sequential deep learning models are the most widely used, followed by convolutional models and traditional machine learning models, which appear in roughly equal proportions. The Bayesian network family is also common. Meanwhile, transformers and GNNs models are more recent and less frequently used compared to sequential and convolutional models, but their adoption is increasing. Figure 18 presents top 10 model usage trends over time. Sequential models, saw an increase beginning in 2018, peaking in 2022, followed by a slight decline. In contrast, GNNs and transformers began appearing in 2021 and 2022 respectively, and they continue to rise till 2025, specifically the transformers which almost keep their usage rate. Graph representation learning (which includes KGEs) and Bayesian inference models also have an influence in 2025, reflecting a growing interest in interpretable and structured models. Convolutional models increased from 2017 and peaked in 2022, then began to decline. Figure 19 shows how model families are distributed across prediction tasks (intention, trajectory, joint (Both)) and vehicle focus (ego, surrounding). From this figure, it can be observed that:

- Sequential models dominate in trajectory prediction for surrounding vehicles.
- Convolutional models are frequently used in intention prediction for surrounding vehicles.
- Traditional ML and Bayesian networks appear mostly in intention prediction, covering both ego and surrounding vehicles.

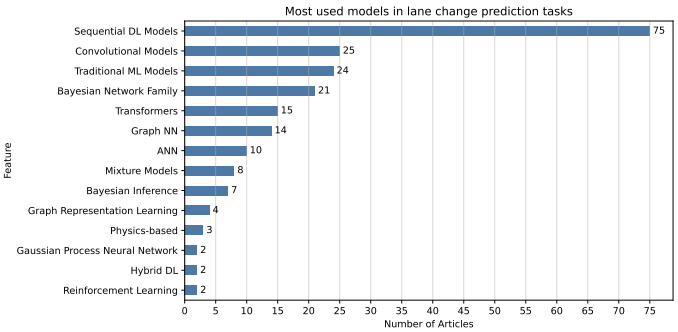

Fig. 17: Most frequently used models in lane-change prediction tasks. Sequential deep learning models (RNNs, LSTMs, GRUs) dominate the field, followed by CNNs, traditional ML models, and Bayesian networks. Emerging approaches, such as GNNs and Transformers, are also gaining momentum, although they remain less commonly applied compared to conventional methods. Less commonly used approaches include hybrid deep learning models, reinforcement learning, and physics-based models.

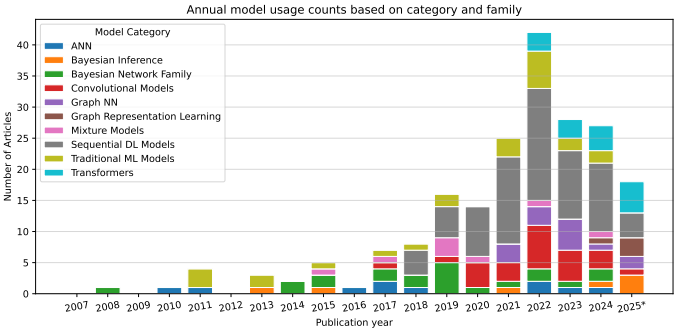

Fig. 18: Stacked bar chart shows annual distribution of model usage in lane-change prediction studies by model family and class. Sequential DL models have seen a sharp rise since 2018, peaking in 2022. Transformer and GNN-based models gained attraction more recently, while traditional ML and Bayesian approaches were more common in earlier years.

- GNNs are focused on trajectory prediction for surrounding vehicles.
- Transformers are used across both intention and trajectory prediction tasks for surrounding vehicles but remain not used for ego vehicle trajectory and joint prediction task.
- KGEs and Bayesian inference are used primarily for intention prediction of surrounding vehicles.

Figure 20 offers a heatmap of specific model usage across the surveyed works. The heatmap shows that LSTM is the most widely used model overall, especially in trajectory prediction for surrounding vehicles. Variants of the used LSTMs include: attention-based LSTMs [10, 84, 107, 108, 140, 145], encoder-decoder LSTMs [21, 22, 35, 45–47, 58, 148, 153], attention-based encoder-decoder LSTMs [33, 34, 56, 130, 134, 135, 141, 143, 144, 152, 159, 160, 163], bidirectional LSTMs [8, 68, 164], and attention-based bidirectional LSTMs [70, 71]. Also, CNNs are the most used convolutional model; TCNs are less common. Transformers are used in intention prediction tasks for both vehicle types and in some trajectory and joint prediction tasks for surrounding vehicles. HMMs are used primarily for intention prediction, balanced across ego and surrounding vehicles. SVM is the most used model across all traditional ML models and it is popular for intention prediction tasks. GCNs are the most commonly used GNNs and are focused on trajectory prediction for surrounding vehicles. Bayesian inference models are limited to a few recent studies, used mainly for surrounding vehicle intention prediction. At the right edge of the heatmap, several models appear only once or twice. These include Naive Bayes, static Bayesian networks, GMMs, LLMs, fuzzy neural networks, and physics-informed neural networks. Their limited use is due to either the novelty of the model as in LLMs (belongs to transformers family) which may gain more focus in the future, or due to the presence of alternative models (from the left side of the heatmap) that currently demonstrate better performance or broader applicability. Finally, Table V provides the full list of models, their categories, and the article IDs in which they were used. It is important to clarify that many papers use multiple models in combination (e.g., CNN + LSTM). However, in this survey, each model is treated independently for counting and analysis purposes. That is, if an article uses three models, each one is counted separately. The aim is to highlight how frequently each model is used across the literature, not whether they were fused or cascaded in a specific architecture.

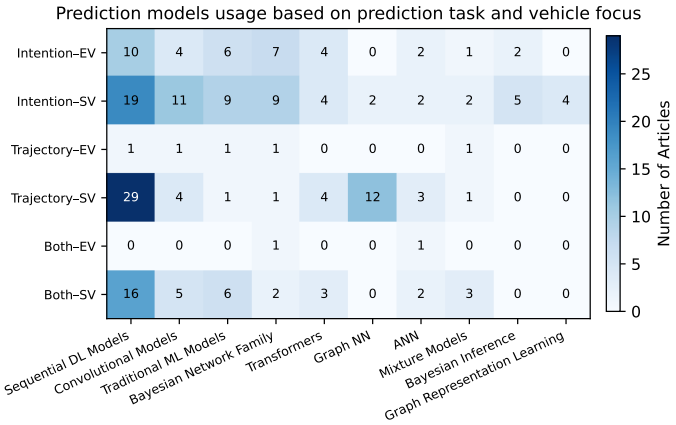

Fig. 19: Heatmap that presents the distribution of model usage by prediction task and vehicle perspective. Sequential deep learning models dominate, especially in trajectory prediction for surrounding vehicles. GNNs are also concentrated in this category but remain largely unexplored in other settings. Even in trajectory prediction tasks, GNNs remain relatively underexplored compared to sequential models. Transformers, despite their popularity in deep learning, are also rarely used compared to sequential models. Traditional ML models and the Bayesian Network Family are more commonly applied in intention prediction tasks.

VII. Performance Evaluation and Prediction Horizons

Evaluating prediction models in the context of lane-change prediction depends on the type of task being addressed. Intention prediction, which classifies whether a vehicle will

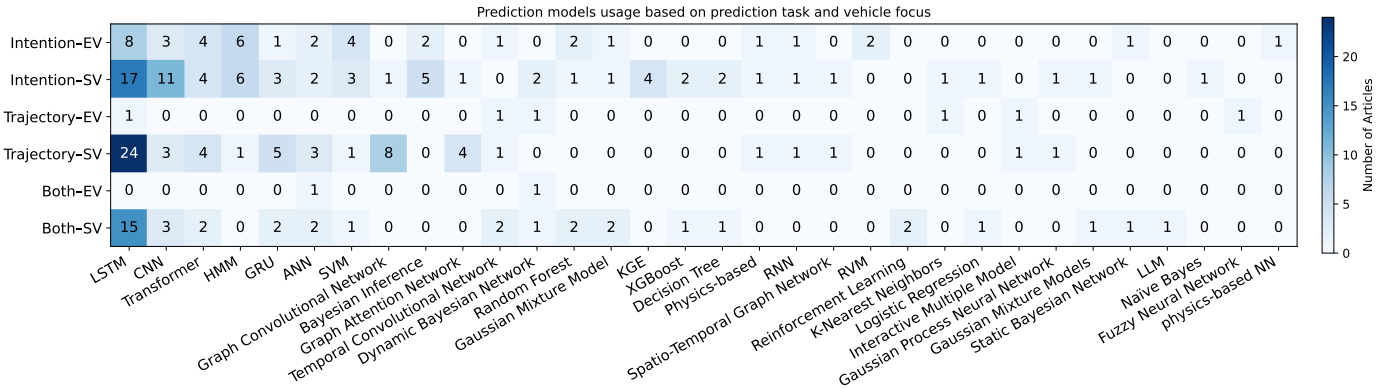

Fig. 20: Heatmap that illustrates the detailed distribution of model categories. Sequential deep learning models dominate overall, with LSTM being the most frequently used variant, particularly in trajectory prediction for surrounding vehicles. CNNs are primarily used in intention prediction tasks, particularly when focusing on surrounding vehicles. Among Graph Neural Networks, Graph Convolutional Networks (GCNs) are the most common, exclusively applied in trajectory prediction for surrounding vehicles, with no instances for ego-vehicle prediction.

keep its lane or perform a lane change, is treated as a multi-class classification problem. In contrast, trajectory prediction, which forecasts future positions of the vehicle, is approached as a regression task. Each of these tasks requires distinct sets of evaluation metrics. The most commonly used metrics for intention prediction include: accuracy, precision, recall, f1-score, Area Under the Curve (AUC), and Receiver Operating Characteristic (ROC). For trajectory prediction, the commonly adopted metrics are: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Final Displacement Error (FDE), Average Displacement Error (ADE), Mean Squared Error (MSE), Negative Log-Likelihood (NLL), and Absolute Error (AE). Figure 21 summarizes the frequency with which these metrics appear in the surveyed literature. In addition to the

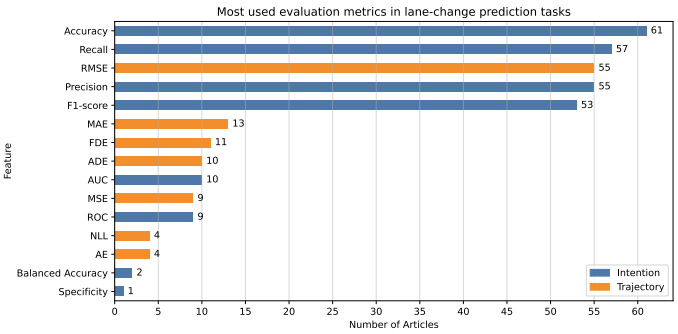

Fig. 21: Most used evaluation metrics in lane change prediction tasks. Classification-based metrics such as Accuracy (61 papers), Recall (57), Precision (55), and F1-score (53) dominate and are predominantly employed in intention prediction tasks. In contrast, regression-based metrics such as RMSE (55 papers), MAE, FDE, and ADE are mainly used in trajectory prediction studies. Notably, RMSE is the most popular among trajectory-focused works. The observed trend reflects a clear division in metric preferences for each prediction task type.

type of metric used, another important factor in evaluating prediction models is the prediction horizon. The choice of horizon length has a strong impact on model design, accuracy, and practical usability. Figure 22 shows the distribution of prediction horizons across the surveyed literature, segmented by prediction task and vehicle focus. For intention prediction, especially when applied to the ego vehicle, most models' prediction horizons are from 2 to 3 seconds. Surrounding vehicle intention prediction follow a similar trend. For trajectory prediction of surrounding vehicles, the standard prediction horizon is typically longer around 5 seconds. When it comes to joint prediction tasks, the final output is often a trajectory, and thus a 5-second prediction horizon is commonly generated. In such cases, the predicted intention serves as an intermediate input for trajectory prediction.

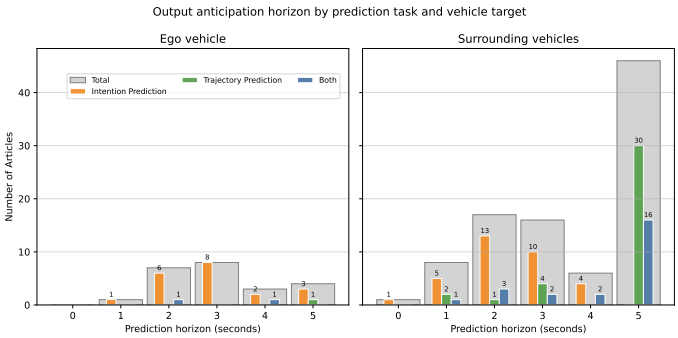

Fig. 22: Distribution of prediction horizons across vehicle focus and prediction task. Prediction horizons (2-3 seconds) dominate across intention prediction, and 5 seconds for trajectory and joint predictions of surrounding vehicles.

VIII. Discussion and Open Challenges

A. Interpretability and Explainability of Models

Most of the research in this survey focuses on getting the best possible results using different evaluation scores for different prediction tasks. However, there is a clear gap when it comes to discussing other important aspects of the models, such as how far these models are understandable and explainable. These aspects include interpretability, transparency, and explainability, which are often not given enough attention.

The interpretability term can mean different things in different studies. Some define it as how one feature affects certain

TABLE V: Categorization of the surveyed studies by model family and specific model type, along with corresponding article IDs.

Model Family	Specific Model	Article ID
Sequential Models	LSTM	[8–10, 12, 21, 22, 26, 27, 29, 32–35, 37, 39, 43, 45–47, 53, 54, 56, 58, 68–73, 84, 86–89, 92, 94, 98–100, 106–108, 117, 124, 130, 134–136, 139–141, 143–145, 147, 148, 150, 152–154, 156, 159, 160, 163, 164]
Sequential Models	GRU	[9, 36, 43, 81, 101, 125, 130, 137, 155, 158, 162]
Sequential Models	RNN	[80, 88, 157]
Convolutional Models	CNN	[38, 39, 47, 68, 74, 81, 85, 87, 99, 103, 106, 110–112, 118, 123, 127, 132, 133, 142]
Convolutional Models	TCN	[12, 42, 54, 69, 155],
Traditional ML Models	SVM	[25, 44, 49, 64, 66, 72, 79, 90, 97, 120]
Traditional ML Models	Random Forest	[28, 35, 72, 79, 97]
Traditional ML Models	XGBoost	[29, 96, 101]
Traditional ML Models	Decision Tree	[10, 31, 99]
Traditional ML Models	RVM	[61, 65]
Traditional ML Models	KNN	[20, 94]
Traditional ML Models	Logistic Regression	[51, 102]
Bayesian Networks	DBN HMM	[17, 19, 30, 41, 57, 59, 62, 79, 82, 91, 94, 105, 114–116, 122, 129, 138]
Bayesian Networks	SBN	[60] [51]
Bayesian Networks	Naive Bayes	[83]
GNN	GCN	[104, 136, 137, 147, 151, 153, 156, 158, 167]
GNN	GAN	[104, 157, 165–167]
GNN	ST-GNN	[108, 168]
Transformer	-	[11, 40, 67, 76–78, 95, 104, 119, 125, 142, 149, 165, 168]
ANN	-	[23, 24, 42, 49, 52, 66, 79, 90, 93, 146]
Mixture Models	GMM	[28, 31, 44, 60, 116, 126]
Mixture Models	IMM	[129, 131]
Bayesian Inference	-	[13, 14, 18, 62, 64, 92, 113]
Graph Representation Learning	KGE	[13, 14, 18, 113]

stages inside the model. Others see it as which inputs have more influence on the model's output. However, in this survey, we define interpretability in a different manner: it refers to how easily a person can understand why the model produced a specific result based on the input. For example, a model receives numerical inputs, such as a lateral velocity of 0 m/s and a time-to-collision of 4 seconds, and it predicts a left lane change. In another case, the input is linguistic, stating that the vehicle is moving straight. The time to collision is marked as high risk, and the model predicts a left lane change. The second example is more interpretable. It's easier for a human to follow up and understand why the model made that prediction, and that is because the inputs are more understandable. So, interpretability means making the input-output relationship easier to follow for users

Moving on to transparency, this refers to how traceable and understandable the internal workings of a model are. In other words, can we determine how specific inputs lead to the final output? This is addressed in some articles, but it is discussed under the umbrella of interpretability. Here, this will be discussed from the perspective of transparency. Models such as LSTMs, GRUs, and CNNs are often considered "black boxes" because their internal decision-making processes are not easily accessible or explainable. Some studies have attempted to address this by incorporating attention mechanisms (such as attention-based LSTMs or transformers) to highlight influential inputs. However, this doesn't fully solve the issue. Such models function as "gray boxes": they offer partial visibility, indicating which features influence the output, but do not fully reveal the decision-making pathway from input to output. Full traceability is still challenging to achieve. Some recent efforts have explored KGEs and Bayesian inference [13, 14, 18, 113]

to improve transparency. Bayesian inference offers a degree of traceability by modeling probabilistic relationships. However, even these models are not fully transparent because the KGE component functions as a black box, making it unclear how the input is processed before the KGE output is passed to the Bayesian inference stage.

Finally, explainability is defined as the ability of the model to generate natural-language justifications for its predictions, providing human-understandable reasons behind the predicted maneuver or trajectory. A truly explainable model can, for instance, output a sentence like: "The vehicle intends to change lanes to the left due to high collision risk with the leading vehicle." Such output helps build user trust and makes the model's decisions more justifiable.

In summary, while many models optimize performance metrics, a significant research gap remains in ensuring that predictions are interpretable, transparent, and explainable to end users. Addressing this gap is essential for building trust in predictive systems for autonomous driving.

As shown in Figure 23, most of the reviewed models are not interpretable. Only five articles [13, 14, 18, 55, 113] used models where it is possible to understand the inputs and outputs easily. In terms of transparency, only about 11% [13, 51, 56, 69, 70, 78, 83, 84, 96, 107, 113, 125, 131, 134, 135, 140, 166–168] of the models state that they provide visibility into their internal processing. But even these do not offer full visibility. Instead, they provide what could be called semi-transparency. For example, some models show which input features were most influential, either through attention weights or feature importance techniques. However, they still don't show how these features are processed internally. It's like looking through a small hole into a closed box, you see a part of what's happening, but not everything. Other studies offer slightly more detail such as the model that combines KGEs with Bayesian inference. In such pipeline, the transparency begins at the stage where KGE modules produce triple-based probabilities, and the downstream inference through Bayesian reasoning becomes traceable. However, the part before this (how the knowledge graph generates those triples' scores) is still hidden meaning that the overall model still cannot be considered fully transparent because it is partially visible, it can be considered as a grey-box. In terms of explainability, only three articles generate natural language explanations alongside predictions. Two of them [13, 113] adopt Retrieval-Augmented Generation (RAG), in which the model retrieves relevant documents that resemble the current driving context and generates explanations grounded in these external references. This improves transparency by associating the prediction with verifiable knowledge and helps build user trust. The third article [55] fine-tunes a Large Language Model (LLM) to generate explanations using chain-of-thought reasoning. In this setup, the LLM not only predicts future maneuvers but also states a step-by-step reasoning for its prediction. However, since these explanations are produced based on internal latent representations learned during fine-tuning, the internal workings of the model remain inaccessible. Furthermore, LLMs still have access to general knowledge acquired before fine-tuning, which raises questions about consistency and transparency. Compared to LLM-based approaches, RAG offers more controlled and grounded outputs because it draws explicitly from the provided retrieval documents. A detailed summary of each article's level of interpretability, semi-transparency, and explainability is provided in Table VI. It highlights that there are many semi-transparent models, yet remain non-interpretable and non-explainable. Most of these models rely on attention mechanisms for transparency. Only a small subset of works achieve any form of interpretability, and even fewer deliver full natural language explanations, emphasizing a significant research gap that deserves attention.

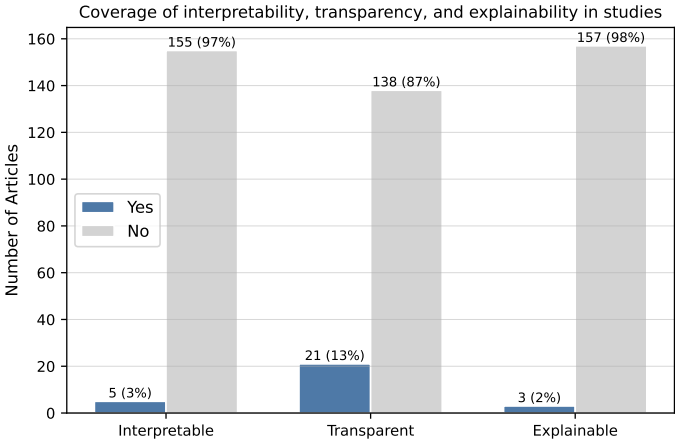

Fig. 23: Bar chart that illustrates the limited attention given to interpretability, transparency, and explainability in lane-change prediction studies. Only 3% of the reviewed works explicitly employ interpretable models, while transparency is addressed in 11% of cases. Notably, only 2% of the studies provide natural language explainable outputs that help users understand the rationale behind the predictions.

B. Decision Making and Real-World Testing

A prediction only becomes useful when it leads to a decision. However, despite the rich diversity in modeling approaches for predicting lane-change intentions or trajectories, most studies treat prediction as an isolated module, without considering the vehicle's decision that should be taken based on the prediction to avoid potential risks posed by surrounding vehicles. Only 4% [9, 13, 14, 85, 107, 127, 159] of the reviewed articles connect their prediction models to any form of decision-making. This leaves a major practical gap: predicting intention or trajectory only is not enough unless the vehicle acts accordingly, and most of the studies stop short at this crucial step. Furthermore, real-world validation is extremely rare. As shown in Figure 24 and summarized in Table VI, only 3% of the articles [14, 61, 65, 127, 159], conducted real-world testing, and 6% [9, 13, 35, 58, 60, 70, 74, 76, 85] included simulation-based validation that goes beyond numerical evaluation. The vast majority (91%) relied solely on quantitative metrics (e.g., accuracy, f1-score, RMSE) without implementing the prediction in a physical or simulation based settings. These studies validate only at the level of output prediction accuracy, missing the opportunity to assess the actual system behavior under dynamic driving

conditions. Columns five and six in Table VI present how the reviewed studies align across decision-making integration and validation type. Some articles rely solely on simulation-based validation, such as [35, 58, 60, 70, 74, 76]. Others focus only on hardware implementation without incorporating any decision-making component [61, 65]. A third group includes studies that integrate decision-making with simulation-based validation [9, 13, 85], while another subset combines decision-making with hardware implementation [14, 127, 159]. The remaining studies neither include decision-making nor hardware or simulation-based validation. Instead, they rely purely on numerical results derived from performance metrics. From this table, it can be concluded that most research continues to treat prediction as an isolated module. Very few works address how predictions influence vehicle actions, and even fewer combine this with interpretability or explainability. Bridging this gap between prediction and decision-making, especially through transparent and explainable models, remains an open and potential direction for future developments.

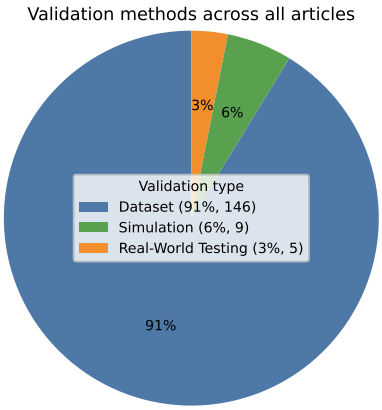

Fig. 24: This figure presents the distribution of validation approaches across all reviewed articles. A vast majority (91%, 147 articles) rely solely on public or collected datasets. Simulation-based validation is applied in only 6% (9 articles), while real-world testing is conducted in just 3% (5 articles). This highlights a strong dependence on offline evaluation and limitations in bridging the gap between prediction models and practical deployment.

C. Summary of Gaps

Through the systematic review of lane change prediction literature, several recurring research gaps have been identified that warrant further investigation:

1) **Missing public benchmark with driver-internal cues:** No open dataset combines exterior traffic context with driver-state signals (eye-gaze, head-pose, biometrics). This hinders comparable research on ego-vehicle intention prediction.
2) **Under-use of contextual features in trajectory forecasting:** While kinematic features dominate the input space for trajectory prediction models, contextual factors such as road geometry, lane offset, and traffic semantics remain under-explored.
3) **Limited Diversity of Model Families:** Although DL models especially sequential models are extensively applied, other model families such as GNNs and transformers remain comparatively under-utilised.
4) **Lack of interpretability, transparency & explainability:** Fewer than 5% of studies employ interpretable models; even fewer generate natural-language explanations, making it hard for humans to understand predictions. Also most of current models function as black boxes, with only a small subset offering traceablity or providing insights into which features influence the prediction.
5) **Weak Integration of Prediction with Decision-Making:** The majority of research treats prediction as an isolated module, without connecting it to downstream vehicle actions. Only a limited number of studies incorporate prediction results into decision-making pipelines.
6) **Over-reliance on Offline Metrics:** Most studies rely solely on offline performance metrics such as accuracy or RMSE, without evaluating the model's behavior in interactive settings. Simulation-based validation is rare, and real-world hardware implementations are even less frequent, limiting the practical credibility of many proposed approaches.

These gaps reveal critical avenues for future exploration and highlight the need for more holistic, interpretable, and practically validated approaches in lane change prediction.

IX. Conclusion

This survey has provided a comprehensive and structured analysis of over 150 research articles published between 2007 and 2025, covering both lane change intention and trajectory prediction. In contrast to prior surveys that often focus narrowly on individual prediction tasks or methodologies, this work offers an integrated view of the entire pipeline from sensor configurations and input feature types to modeling techniques and evaluation strategies. By presenting a detailed statistical analysis, this survey sheds light on critical trends and gaps within the literature. Notably, the field remains imbalanced in its focus, with limited research addressing ego vehicle prediction, driver-internal cues, or contextual information. In addition, despite the growing complexity and accuracy of prediction models, concerns around their interpretability, transparency, and explainability are largely unaddressed. The lack of integration between prediction and decision-making, along with rare real-world validation, further limits the practical deployment of these systems. To advance the field, future research should prioritize the development of explainable and transparent models that are not only accurate but also trustworthy. Integrating prediction outputs into downstream decision-making modules and validating these systems through hardware-in-the-loop testing or real-world trials will be essential for practical deployment. Furthermore, creating benchmark datasets that combine external traffic observations with internal driver states can support more robust and generalizable models. Finally, addressing these gaps will be key to building lane change prediction systems that are not only technically sound but also interpretable, interactive, and safe for real-world use in autonomous driving.

TABLE VI: Overview of related studies categorized based on interpretability, semi-transparency, explainability, decision-making integration, and validation type.

Work	Interpretability	Semi-Transparency	Explainability	Decision Making	Validation Type
[35, 58, 60, 74, 76]	✗	✗	✗	✗	Simulation
[61, 65]	✗	✗	✗	✗	Hardware
[9, 85]	✗	✗	✗	✓	Simulation
[127, 159]	✗	✗	✗	✓	Hardware
[10, 51, 56, 69, 78, 83, 84, 96, 125, 131, 134, 135, 140, 166–168]	✗	✓	✗	✗	Dataset
[70]	✗	✓	✗	✗	Simulation
[107]	✗	✓	✗	✓	Dataset
[55]	✓	✗	✓	✗	Dataset
[14]	✓	✗	✗	✓	Hardware
[14, 18]	✓	✓	✗	✗	Dataset
[113]	✓	✓	✓	✗	Dataset
[13]	✓	✓	✓	✓	Simulation

Acknowledgment

This research has been funded by the HEIDI project of the European Commission under Grant Agreement: 101069538.

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